

Online Theory Supplement to *Automation Risk and the Public Balance Sheet*

Nathan Barnard

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Guide to this supplement

The five modules answer technical questions that arise naturally from the article but are not needed to follow its central argument. They map the boundaries of the production branch, the commitment and feasibility convention, local capital–tax dynamics, richer public payoff menus, and the accounting closure behind the two-successor application.

Module	Expert question	Relation to the article
S1	When is the smooth task-production representation valid?	States the global task assignment and the boundaries of the maintained specialization branch.
S2	What makes commitment recursive, and why is a desired position not debt capacity?	Gives the state, restriction–pasting, and viability conditions behind the conditional recursion.
S3	How does the local capital–tax approximation separate Riccati, Sylvester, and Lya-punov blocks?	Supplies a local propagation benchmark rather than a global policy solution.
S4	How do richer public payoff menus and bounds change the projection argument?	Extends the asset menu and its KKT geometry.
S5	Which equations close the two-successor marked-Poisson application?	Records the event accounting and conditional recovery map behind the rank result.

Notation shared with the article is retained. Thus N is Brownian public financial net worth, s is a dollar holding of the external risky claim, and J is fiscal- resource value rather than pledgeable debt capacity. The marked-Poisson module uses its own lower-case variables (e, ψ) for reduced public wealth and public equity exposure; they are not Brownian (N, s) .

1 Task allocation and the specialization boundaries

The article works on the branch in which capital performs automation-intensive tasks and labour performs new tasks. This module states the inequalities that make that branch an optimizer of the underlying assignment problem.

For fixed (z, α, K, L) , let the firm allocate capital and labour between an automation-intensive

composite A and a new-task composite M :

$$y_A = e^{\nu_A} K_A + L_A, \quad y_M = e^{-\nu_N} K_M + L_M, \quad (1.1)$$

$$Y = \max e^z \left(\frac{y_A}{\alpha} \right)^\alpha \left(\frac{y_M}{1-\alpha} \right)^{1-\alpha}, \quad (1.2)$$

subject to nonnegative assignments and $K_A + K_M = K$, $L_A + L_M = L$. With

$$s_A = \log(K/L) - \text{logit}(\alpha) + \nu_A, \quad s_N = \nu_N - \log(K/L) + \text{logit}(\alpha),$$

the optimized production function is

$$Y = \begin{cases} e^z (e^{\nu_A} K + L), & s_A \leq 0, \\ e^{z+\nu_A\alpha} \alpha^{-\alpha} (1-\alpha)^{-(1-\alpha)} K^\alpha L^{1-\alpha}, & s_A \geq 0, \ s_N \geq 0, \\ e^{z+\nu_A\alpha-\nu_N(1-\alpha)} (K + e^{\nu_N} L), & s_N \leq 0. \end{cases} \quad (1.3)$$

The middle expression is strict precisely when $s_A > 0$ and $s_N > 0$. It yields $R^K K = \alpha Y$ and $wL = (1-\alpha)Y$, or equivalently $R^K/e^{\nu_A} < w < e^{\nu_N} R^K$. At either equality adjacent expressions agree but the smooth branch is no longer strict. Derivatives that use the Cobb–Douglas formula are branch derivatives rather than global comparative statics.

For $\log(K/L) \in [k_{\min}, k_{\max}]$ and $\alpha \in [\alpha_{\min}, \alpha_{\max}]$, full specialization holds throughout the rectangle if

$$k_{\min} > -\nu_A + \text{logit}(\alpha_{\max}), \quad k_{\max} < \nu_N + \text{logit}(\alpha_{\min}).$$

Crossing the minimum slack changes the production formula and calls for a separate equilibrium and policy analysis; it does not by itself determine whether the public claim is aligned with worker resources.

2 Commitment, admissibility, and fiscal feasibility

The recursive Brownian representation is a conditional representation of a date-zero commitment plan. Let $S = (z, x, K, N, \tau)$ and let $\mathcal{A}(S)$ denote policies with positive worker consumption, the transfer floor, portfolio bounds, the tax ceiling and inward speed there, integrability, and conditional no-Ponzi, transversality, and continuation-feasibility restrictions. The interior HJB is used only where these restrictions are slack.

All consequences of the past relevant for a future feasible tail must be carried. Write the inherited augmented state as (S_0, M_0) , where M_0 contains surviving promises, multipliers, protected claims, or institutional commitments. In the maintained Brownian closure workers have no wealth or Euler equation, firms are static, external prices are Markov, K and τ are inherited, and prior public trading is summarized by N . Thus M_t is empty only under the stated state-based continuation convention. A protected continuation utility, date-zero residual budget, or history-specific entitlement belongs in M_t and prevents this reduction until its update is specified.

For a bounded stopping time ϑ , restriction takes an admissible date-zero tail and views it as a continuation from $(S_\vartheta, M_\vartheta)$. Pasting joins a feasible prefix to a measurable admissible continuation selected at that state. If controlled laws are well posed, conditional tails integrable, and $\mathcal{A}$ stable under both operations, then

$$V_\vartheta^{\text{commit}}(S_\vartheta, M_\vartheta) = V^{\text{rec}}(S_\vartheta, M_\vartheta).$$

The equality selects an optimal continuation correspondence. Replaying identical controls additionally requires uniqueness and is not implied by equality of values.

Fiscal feasibility is more restrictive than a desired portfolio. In the Brownian economy the residual safe position is $N - s$, with a representative gross-position restriction $-\bar{S}^G \leq s \leq N + \bar{B}^G$. Even when the small-risk demand $s^0 = X - h_J$ is positive, $X = N + J$ includes future worker wages and tax resources: J is not collateral. Feasibility additionally requires a nonempty position interval, a path inside the conditional solvency set, and the stated terminal convention. Thus neither $X > 0$ nor an expected-return calculation characterizes maximum debt capacity.

The jump application sharpens the distinction. With public equity Θ , gross safe debt $B \geq 0$, and net wealth $F = \Theta - B$, an equal increase in equity and debt leaves F unchanged. It changes jump exposure, not current net resources. A strict safe promise instead requires viability after every supported successor; that is a pathwise condition, not a physical-expectation or SDF present-value inequality.

3 Local capital–tax dynamics

This section is a local benchmark around the deterministic interior anchor. It explains how the tax’s forward-looking effect propagates through capital; it does not determine a global nonlinear policy or whether a path remains away from the constraints in Section 2.

Let $y = (z - \bar{z}, x - \bar{x}, k, t)^\top$, where $k = \log(K/\bar{K})$ and $t = \tau - 1/2$. The linearized law is $\dot{y} = Ay + B\nu$, with

$$A = \begin{pmatrix} -\kappa_z & 0 & 0 & 0 \\ 0 & -\kappa_x & 0 & 0 \\ \bar{R}/2 & \bar{R}_x/2 & -\gamma & -2\rho \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = (0, 0, 0, 1)^\top, \quad \gamma = (1 - \bar{\alpha})\bar{R}/2.$$

Partition $y = (e^\top, r^\top)^\top$ into exogenous states e and inherited capital–tax states $r = (k, t)^\top$. For the quadratic fiscal-resource approximation

$$\mathfrak{J}(y) = \bar{\mathfrak{J}} + j^\top y + \frac{1}{2} y^\top H y + o(\|y\|^2),$$

the linear coefficient solves $(\rho I - A^\top)j = f_y - \bar{\mathfrak{J}}d_r$ and curvature solves the discounted Riccati equation

$$0 = Q + A^\top H + H A - \rho H + \chi H B B^\top H, \quad \chi = \bar{\alpha}\kappa_\tau/\bar{R} > 0.$$

Because exogenous shocks do not respond to domestic controls, the nonlinear part is only the 2×2 rr block:

$$0 = Q_{rr} + A_r^\top H_{rr} + H_{rr} A_r - \rho H_{rr} + \chi H_{rr} b b^\top H_{rr}, \quad A_r = \begin{pmatrix} -\gamma & -2\rho \\ 0 & 0 \end{pmatrix}, \quad b = (0, 1)^\top,$$

with $Q_{rr} = \begin{pmatrix} \rho & 2\rho \\ 2\rho & 0 \end{pmatrix}$. After selecting its stabilizing symmetric solution, the cross block H_{er} solves a linear Sylvester equation; substituting it leaves a discounted Lyapunov equation for H_{ee} . The ordering first solves capital–tax feedback, then loads technology shocks onto it.

For $a_0 = \gamma + \rho/2$ and $b_0 = \rho/2$, the intrinsic Hamiltonian polynomial is

$$p(\lambda) = \lambda^4 - (a_0^2 + b_0^2)\lambda^2 + a_0^2 b_0^2 + 8\chi\rho^2\gamma.$$

Under the maintained positive parameters, strict Popov positivity and controllability select the real symmetric stabilizing solution and a Hurwitz closed loop. Its intrinsic stable modes are complex exactly when

$$\chi > \frac{\gamma(\gamma + \rho)^2}{32\rho^2},$$

or, under the primitive mapping,

$$16\kappa_\tau\bar{\alpha}(1 - \bar{\alpha})\rho^2 > \gamma^2(\gamma + \rho)^2.$$

Above this threshold generic homogeneous capital–tax responses have damped oscillatory components. Below it intrinsic roots are real, but coordinate responses can still overshoot through eigenvector mixing or shock forcing. The distinction is local and does not predict a transition after a constraint or production boundary is reached.

4 Alternative public payoff menus and constrained variants

The article’s one-claim geometry is deliberately narrow. Let $\Sigma \in \mathbb{R}^{d \times m}$ have columns equal to the diffusion loadings of m pre-positioned Brownian claims, let $q \in \mathbb{R}^m$ be dollar holdings, and let $\zeta_J \in \mathbb{R}^d$ be fiscal-resource exposure. In the small-risk benchmark, the hedgeable component is

$$P_\Sigma \zeta_J = \Sigma(\Sigma^\top \Sigma)^\dagger \Sigma^\top \zeta_J, \quad \zeta_J^\perp = (I - P_\Sigma)\zeta_J,$$

where the dagger permits redundant claims. A new claim changes span only if it changes $\text{col}(\Sigma)$; a proportional clone can change prices or constraints but cannot remove $\zeta_J^\perp$ in this local geometry.

For a full-column-rank menu and a common deterministic baseline, the claim-equivalent hedge is $h_\Sigma = (\Sigma^\top \Sigma)^{-1} \Sigma^\top \zeta_J$. The article’s scalar formula $s^0 = X - h_J$ is its one-claim counterpart. The welfare comparison attached to completion remains order two between feasible nested menus, holding the deterministic allocation and world prices fixed; it does not complete household markets.

The exact nonlinear control condition is a KKT system. If the portfolio constraint set is $\mathcal{Q} = \{q : Gq \leq g\}$ and $\mathcal{H}(q)$ is the local Hamiltonian block, a candidate satisfies

$$\nabla_q \mathcal{H}(q) - G^\top \mu = 0, \quad \mu \geq 0, \quad Gq - g \leq 0, \quad \mu_i(Gq - g)_i = 0.$$

For one claim this reduces to the article’s signed residual: at an upper bound the marginal valuation wedge in the increasing- s direction is weakly positive, and at a lower bound it is weakly negative. With several claims, only feasible directions receive that interpretation. A capacity shadow value is therefore not an unspanned residual.

A full-return tax can carry a jump payoff when its base includes a capital gain; a predictable tax on rental flows does not. A precommitted mark-indexed schedule is also a richer contingent instrument than a scalar predictable flow rate. Both are different institutional models rather than hidden additions to the article’s asset menu.

5 Marked-Poisson successor closure and event accounting

This module gives the accounting architecture behind the article’s two-successor application. It preserves a small-open economy with an exogenous world SDF. A mark $j \in \{P, F\}$ arrives at a totally inaccessible time with physical intensity λ_j and world risk-neutral intensity

$$\lambda_j^* = \lambda_j M_j^{w,+} / M_0^{w,-}.$$

Physical and risk-neutral intensities have different jobs. Installed-capital equity $Q = qK$ has proportional total-gain jump $\Delta_j^E = (q_j^+ - q)/q$. The predictable source tax is levied on rental-income flows; it has no singular event component and is not selected after observing j .

Use efficiency-unit public wealth $f = F/A$, public equity $\theta = \Theta/A$, and gross safe debt $b = B/A$, so $f = \theta - b$. The reduced variables

$$e = f - qk, \quad \psi = \theta - qk, \quad b = \psi - e$$

separate public net resources from gross exposure. With $\Lambda = \sum_j \lambda_j^* \Delta_j^E$, the pre-arrival consolidated law and successor map are

$$\dot{e} = \bar{r}_0 e + y_0(k) - \iota k - c^W - \psi \Lambda, \quad (5.1)$$

$$e_j^+ = e + \psi \Delta_j^E. \quad (5.2)$$

Tax revenue disappears from the consolidated e law only after consolidation. It still changes investment, q , the total-gain vector, and the decentralizing user-cost wedge. Private owner wealth a obeys $c^K = \rho a$ and

$$\dot{a} = (\bar{r}_0 - \rho - \pi \Lambda) a, \quad a_j^+ = a(1 + \pi \Delta_j^E),$$

where π is the owner's equity share. Together with the current-account identity, these laws replace a closed-economy resource constraint.

For fixed (e, k, a) and a fixed continuation-policy map, (5.2) sends one scalar ψ into the line

$$(e_P^+ - e, e_F^+ - e)^\top = \psi(\Delta_P^E, \Delta_F^E)^\top.$$

The safe asset has no event payoff and the predictable rental-flow tax adds no second event atom. Admissibility still requires $b \geq 0$, positive worker and successor resources, private solvency, protected claims, tax and transfer bounds, and successor-by-successor fiscal viability.

Under maintained one-time protection, the candidate recursive physical state is $S = (s, k, e, a)$, with no residual promise only if old future policy paths are not enforceable entitlements. Restriction and measurable pasting also require a common measurable price and branch selection. If a continuation value or legal claim survives by history, the state is (S, M) ; if q is not selected by S , a price or branch label must be carried as well.

Remaining instruments can be recovered from world equity pricing, the worker transfer identity $t = c^W - w_0(k)$, and $b = \psi - e$. This recovery is admissible only when those objects satisfy the constraints on every relevant continuation. The equations above do not by themselves supply existence, a welfare ranking, or a source-tax conclusion.